

Project Summary: House Price Prediction

Internship Project — Week 1 Prepared by: chappidi Lavanya

1. Project Overview

The objective of this project was to analyze a housing dataset and build a predictive machine learning model to estimate property prices based on various structural and locational features.

2. Key Insights (What Drives House Prices?)

1) What drives prices most?

Property size (**area**), number of **bathrooms** and having **air conditioning** are the biggest factors behind higher house prices.

2) Data Surprise:

Having a **basement** or **main road** access mattered much less to the final price than expected

3) Business Takeaway:

To maximize property value, focus on increasing the livable square footage (` area `) and adding internal upgrades like air conditioning or extra bathrooms.

3. Model Performance Comparison

Two separate regression models were built and evaluated on an 80/20 train-test split (436 training rows, 109 testing rows):

- **Linear Regression (Winner):** Achieved an R^2 score of **0.6529**, meaning it successfully explains **65.3%** of the variance in housing prices.
- **Random Forest Regressor:** Achieved a lower R^2 score of **0.6119**.

Why did Linear Regression win? Housing features naturally scale in a highly predictable, straight-line relationship with price. Because Random Forest is a highly complex model, it struggled to map out its deep "decision trees" on our relatively small dataset (436 rows) and slightly overfit, leading to a lower test score.

4. Business Recommendations

For a real estate firm looking to maximize property portfolio value, resources should be prioritized toward:

1. Optimizing livable square footage (area) during development.
2. Prioritizing high-ROI internal upgrades, specifically adding extra bathrooms and installing central air conditioning, as these features show the strongest direct link to premium price